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Norharmane prototropism in choline chloride-based deep eutectic solvents

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ABSTRACT

Deep eutectic solvents (DESs) are gaining widespread attention as novel and alternate solvents in chemical and biological sciences. Prototropic equilibria are known to be governed by the solubilizing milieu, and thus, it is imperative to study them within DESs. Prototropic behavior of β-carboline commonly named as norharmane (9H-pyrido[3,4-b]indole) is investigated in eight different DESs constituted of the salt choline chloride with eight different H-bond donors (HBDs) to ascertain their role in controlling prototropism within DESs. UV–vis molecular absorbance is used to obtain information on ground-state prototropism whereas steady-state and time-resolved fluorescence are used to investigate prototropic forms in the excited-state. In the ground-state, DESs with HBDs glycerol and ethylene glycol support both neutral and cationic forms of norharmane, however, within DESs constituted of HBDs urea, 1,4-butane-diol, and acetamide, respectively, only the neutral form exists. Within the remaining three DESs with HBDs tetraethylene glycol, 3-phenylpropionic acid, and malonic acid, respectively, only cationic form is supported. As the cationic form is preferred more over the neutral form of norharmane in the excited-state, DESs with HBDs tetraethylene glycol, 3-phenylpropionic acid, and malonic acid, respectively, support only cationic form in the excited-state. DES with glycerol also supports only the cationic form in the excited-state, DES with ethylene glycol, acetamide, and 1,4-butane-diol support both cationic and neutral forms. The DES with urea as the HBD is the only DES that supports only the neutral form in the exited-state. Excited-state intensity decay data confirms the presence of different prototropic forms of norharmane within investigated DESs with distinct decay times for the neutral and cationic forms. Correlation between the common empirical solvent polarity parameter, E_T^N, along with Kamlet-Taft parameters, H-bond donating acidity (α) and H-bond accepting basicity (β), of the DESs and the relative presence of the norharmane prototropic forms is established.

1. Introduction

Deep eutectic solvents (DESs) have gained noticeable attention recently because of their possible application potential in various fields of science and technology. Most of the academic and industrial research laboratories are dependent on the conventional organic solvents (or VOCs) which are volatile, virulent, noxious, and can have deleterious effects on both human and the environment. In an effort to limit the use of organic solvents, DESs have come up as neoteric alternative which is environmentally-benign and inexpensive [1–5]. However, DESs have many advantages over traditional ionic liquids, such as, ease of preparation, low cost, and easy availability from inexpensive components (the components themselves are biodegradable and toxicologically well-characterized) [6–12]. The preparation of an important class of DES involves the combination of quaternary ammonium salt (hydrogen bond acceptor, HBA) with a metal salt or hydrogen bond donor (HBD). The charge dispersion occurring via hydrogen bond interaction between HBD and HBA, for example, halide component of HBA and the HBD sites is responsible for the depression in the melting point of the resultant mixture relative to the melting points of their individual components [13]. DESs offer utility in various fields as chemical and physical properties of these solvents can be modified just by substituting the different components that can be tailored to the desired chemistry [14–18]. DESs are renowned with a number of intriguing physical and chemical properties, such as low volatility, wide liquid-range, and non-flammability. Hence, they are attracting the interest of many researchers from various interdisciplinary areas [19–23]. However, it becomes mandatory to mention that not all the DESs have the property of negligible non-flammability relative to other organic solvents over a wide range of temperature. The constituting species and their vapor pressure predict the non-flammable nature of DESs. Recently some DESs have been found to have severe evaporation as reported by Mu. et al. [24]. The DES mixture is often assumed to have negligible vapor pressure due to the presence of one of its non-volatile

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components. However, an extensive investigation is mandatory for the justification of this assumption [25–27].

Proton transfer is one of the most fundamental and significant phenomenon, widely occurring in biological and chemical sciences. Subsequently, the investigation of proton transfer processes in interdisciplinary fields of science and technology has gained a major research thrust worldwide. Besides charge separation, mass transfer also occurs during the proton transfer reactions; in general, the process gets more influenced in the excited state as compared to the ground state. Upon excitation of a molecule, redistribution of electron density usually occurs which enhances/diminishes the magnitude of dipole moment that further influences the acidic and basic properties in the excited state of the molecule [28,29]. Proton transfer phenomenon in excited electronic states has been primarily explored, among notable others, by the innovative work of Forster [30] and Weller [31]. There are plethora of reports available covering wide aspects of the excited-state proton-transfer (ESPT) process [32]. The importance of ESPT reactions is readily established by their various potential applications, such as, in photodynamic therapy, as laser dyes, as photo-protecting agents, and in fluorescence sensing for biological purposes [33,34].

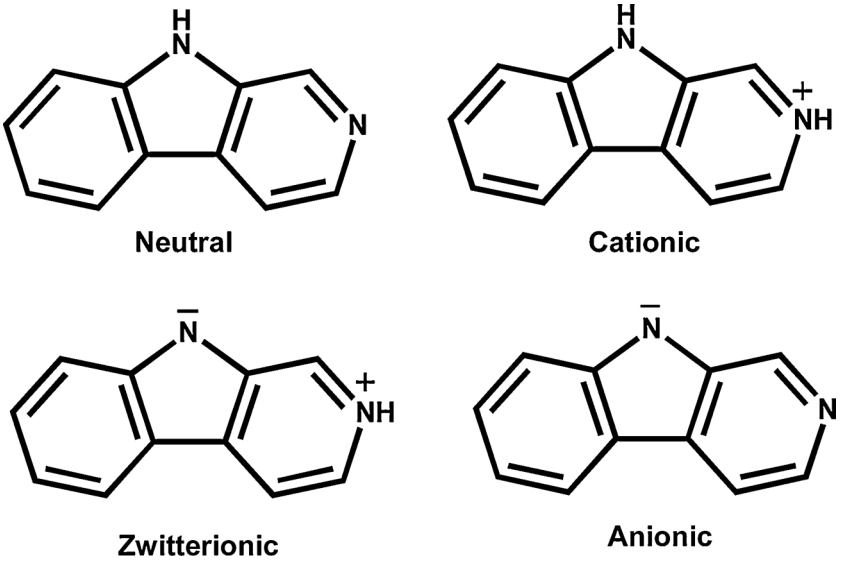
The importance of ESPT in interdisciplinary sciences is unequivocally recognised and have become a topic of intensive research. In this context, β-carboline alkaloids have been studied due to their interesting photophysical and photochemical properties [35,36]. They are also proposed as common fluorophores and photosensitizers in chemistry and biology [37]. β-Carboline commonly termed as norharmane (9H-pyrido[3,4-b]indole) contains a π-deficient pyridinic ring linked to a π-excessive indole ring. Upon excitation, the two different types of nitrogen atoms (pyridine and pyrrole) present in norharmane create a significant and differential charge density separation, which leads to changes in ionization constant for proton abstraction from pyrrolic nitrogen. Consequently, the excited state of this molecule may possibly has different type of prototropic forms. The presence of two functional centres in norharmane generates interesting acid-base properties in the ground as well as the excited state, which further depends on the solvent and pH of the concerned solution [38–43]. Norharmane, in a single molecule, contains both hydrogen bond donor and acceptor characteristics because of the presence of pyrrolic and pyridinic acidic and basic nitrogens. Various prototropic forms of norharmane, such as, cationic, neutral, anionic and zwitterionic, have been proposed in aqueous and organic solvents (Scheme 1) [44–49]. The existence of these prototropic forms in the excited state is influenced by the interactions present between the solvent and norharmane [50]. Hence, the absorption and fluorescence behaviour of norharmane significantly depends on the type of solvent [51–55]. As a result, norharmane has become a molecule of interest in the field of proton transfer and has evoked curiosity among the researchers for further exploring its solvent-dependent photophysical and photochemical properties.

Several studies have been carried out on norharmane prototropism within aqueous and conventional organic solvents, underlining the acid-base properties of norharmane in the excited state [56–63]. Capriati and co-workers have studied the photochemical reactions and reported the stabilized charge-separated species within DES [64]. Therefore, DESs are realized to have the ability to stabilize the charged species formed during the span of a photochemical reaction. In the present study, we have explored DESs as non-aqueous solubilizing media for norharmane prototropism. Interaction of HBD component of DES with norharmane facilitates the formation of different prototropic species in the ground as well as the excited state. Current study shows how HBD interacts with norharmane by keeping the HBA of DES as a non-variant component. As far as we know, this is the first report underlying the norharmane prototropism within DESs. In the present investigation, we have judiciously selected eight DESs having the same HBA component, the salt choline chloride (ChCl), and HBDs, possessing different functionalities and properties. DESs prepared are: URCC (urea + ChCl in 2 : 1 mol ratio), AACC (acetamide + ChCl in 2 : 1 mol ratio), EGCC (ethylene glycol + ChCl in 2 : 1 mol ratio), GLCC (glycerol + ChCl in 2 : 1 mol ratio), TGCC (tetraethylene glycol + ChCl in 2 : 1 mol ratio), PACC (3-phenylpropionic acid + ChCl in 2 : 1 mol ratio), MACC (malonic acid + ChCl in 1 : 1 mol ratio) and BDCC (1,4-butanediol + ChCl in 3 : 1 mol ratio)(structure presented in Scheme2). Effect of HBD of a DES on prototropic behavior of norharmane is explored in detail and is analysed and reported in the work.

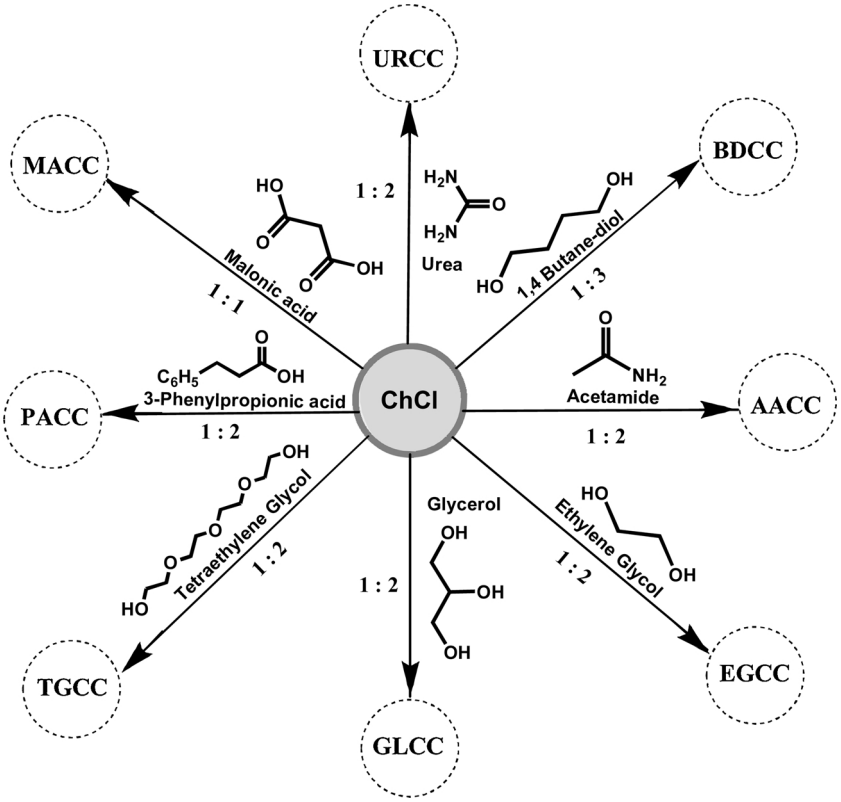
2. Experimental section

2.1. Materials

URCC, GLCC, EGCC, PACC, TGCC and AACC were prepared by mixing ChCl (> 99% from Sigma-Aldrich) with urea (> 99% from Sigma-Aldrich), glycerol (> 99% from Sigma-Aldrich), ethylene glycol (> 99% from Sigma-Aldrich), 3-phenylpropionic acid (> 99% from Sigma-Aldrich), tetraethylene glycol (> 99% from Sigma-Aldrich) and acetamide (> 99% from Sigma-Aldrich), respectively, in 1 : 2 mole ratio followed by stirring under heating ϵ 80 °C) until a homogeneous, colorless liquid was formed. MACC and BDCC were prepared by mixing



Scheme 1. Structures of different prototropic forms of norharmane.



Scheme 2. Compositions of various DESs (as molar ratio of HBA: HBD) used in this study.

ChCl with malonic acid (> 99% from Sigma-Aldrich) and 1,4-butanediol (> 99% from Sigma-Aldrich) in 1 : 1 and 1 : 3 mole ratios, respectively, followed by gentle heating until homogeneous, colorless liquid was obtained. DESs thus prepared were rigorously dried under vacuum for at least 24 h. Karl-Fisher Titrator was subsequently used to measure the water content of the DESs prior to their use. DESs were dried until water content became < 100 ppm [65]. Norharmane was used as received from Sigma-Aldrich. PEGs of average molecular weight 200 (PEG200), 400 (PEG400), and 600 (PEG600) were obtained in the highest purity from Sigma-Aldrich and were used as received. All the organic solvents used — cyclohexane, methanol, trifluoroethanol, butanol, DMSO, and propanol – were purchased in highest purity available from Sigma-Aldrich. 2,6-Dichloro-4-(2,4,6-triphenyl-*N*-pyridino)-phenolate (betaine dye 33), 4-nitroaniline, and *N,N*-diethyl-4-nitroaniline were purchased in the highest available purity from Fluka (≥99%, HPLC), Spectrochem. Co. Ltd., and Frinton Laboratories, respectively. Absolute ethanol was used to prepare probe stock solution.

2.2. Methods

The required amount of the probe norharmane was weighed using an analytical balance with a precision of ± 0.1 mg. Stock solution of norharmane was prepared by dissolving in ethanol in pre-cleaned amber glass vials and stored at 4 ± 1 °C to retard any photochemical reactions. An appropriate amount of the probe solution from the stock was transferred to the 1-cm path length quartz cuvette. Ethanol was evaporated using a gentle stream of high purity nitrogen gas and DES/aqueous solution/organic solvent was added in pre-calculated amount to achieve the desired final concentration of the probe. NaOH and nitric acid were used for maintaining the pH of aqueous (HPLC grade water) system by using Eutech Instruments 700 pH meter.

A Perkin-Elmer Lambda 35 double beam spectrophotometer with variable bandwidth was used for the acquisition of the UV−vis molecular absorption spectra. Steady-state fluorescence spectra were acquired on a Jobin-Yvon Fluorolog-3 (model FL-3-11) modular spectrofluorometer equipped with a 450 W Xe arc lamp as the excitation source and single-grating monochromators as wavelength selection devices with a photomultiplier tube as the detector. All spectra were duly corrected by subtracting the spectral responses from suitable blanks prior to data analysis. Data analysis was performed using SigmaPlot v10 software.

In order to obtain the time-resolved fluorescence emission of norharmane, excited-state intensity decay data were acquired in the time-domain using Horiba-Jobin Yvon, Inc. Fluorocube time-correlated single photon counting (TCSPC) fluorimeter. Norharmane dissolved in URCC, GLCC, EGCC, PACC, MACC and BDCC at room temperature whereas in TGCC and AACC at 55 ± 2 °C was excited using a 340 nm UV-pulsed NanoLED-340 source having pulse width < 1.0 ns. The instrument response width is < 0.2 ns. The emission was collected using Peltier-cooled red-sensitive TBX-04 PMT detection module at 380 nm and/or 450 nm. The data was collected with DAQ-MCA-3 Series (P7882) multichannel analyzer. The instrument response function (IRF) was obtained using a scattering solution of glycogen in water (glycogen from bovine liver, Type IX, Sigma-Aldrich). The excited-state intensity decays were analyzed using DAS6 analysis software and were fitted to the desired decay models.

Kamlet–Taft solvatochromic indicators of solvent dipolarity/polarizability (π*), hydrogen bond donating (HBD) acidity (α), and hydrogen bond accepting (HBA) basicity (β) were also used to gain further insight into our systems. The π* parameters were obtained from the absorption maximum ($\bar{\upsilon}_{DENA}$, in kK) of *N*, *N*-diethyl-4-nitroaniline (DENA), a non-HBD solute, using Eq. (1):

$$\pi^* = 8.649 - 0.314\bar{\upsilon}_{DENA} \quad (1)$$

The E_T (33), E_T(30), E_T^N values were determined using following equations:

$$E_T\,(33) = \frac{28591.5}{\lambda_{max}} \quad (2)$$

$$E_T(30) = 0.9953(\pm 0.0287) \times E_T(33) - 8.1132(\pm 1.6546) \quad (3)$$

$$E_T^N = \frac{[E_T(30)\text{solvent} - 30.7]}{32.4} \quad (4)$$

The α values were determined from $E_T(30)$ and π^* using Eq. (5):

$$\alpha = 0.0649E_T(30) - 2.03 - 0.72\pi^* \quad (5)$$

and β values are calculated from the enhanced solvatochromic shift of NA similar to its homomorph DENA by Eq. (6)

$$\beta = -0.357\bar{\upsilon}_{NA} - 1.176\,\pi^* + 11.12 \quad (6)$$

3. Results and discussion

3.1. UV–vis molecular studies

UV-vis molecular absorption spectra of norharmane dissolved in water at pH 2 is characterized by appearance of a low energy transition at ca. 370 nm along with a higher energy band centered at ca. 300 nm (Fig. S1, panel A). These bands are assigned to the cationic form (C) of norharmane. However, at pH 7, the lower energy bands characterizing the neutral form (N) located at ca. 337 and ca. 347 nm, are clearly visible along with a prominent shoulder at ca. 370 nm implying presence of cationic form along with the neutral at this pH. Only the bands corresponding to the neutral form (ca. 337 nm and ca. 347 nm) are present at pH 12. The pK_a of the neutral to cationic form is reported to be ~7.2 and pK_a of the cationic to zwitterionic (Z) form is ~10.9 [33]. The absorbance transitions of norharmane in aqueous solutions of different pH are listed in Table 1 and are in agreement with the literature reports [37].

Truncated normalized UV–vis absorption spectra of norharmane dissolved in each of the DESs at 298.15 K (except for AACC and TGCC for which the data is collected at 328.15 K) are presented in Fig. 1 (band maxima in the complete UV–vis region for all DESs are listed in Table 1). A careful examination of the data reveals presence of rather interesting features that are governed by the identity of the H-bond donor of the DES. While in GLCC and EGCC, the band with maxima at ~370 nm characterizing the cationic form is clearly dominant, presence of neutral form is also evident by the ~350 nm band (dominant band in EGCC and as a strong shoulder in GLCC). Surprisingly, however, in URCC, BDCC, and AACC, the band at 370 nm is completely absent and the spectrum is characterized by the bands corresponding only to the neutral form of norharmane. For the remaining three DESs, TGCC, PACC, and MACC, the bands corresponding to the neutral form of norharmane are completely absent whereas cationic form is the only form present within these DESs. Different prototropic forms of norharmane in ground-state exist in different DESs; the H-bond donor of the DES clearly controls the prototropism in ground-state.

Table 1
Absorbance (λ_{abs}^{max}) and fluorescence emission (λ_{flu}^{max}) maxima of norharmane (25 μM for (UV–vis absorption and 10 μM for fluorescence) dissolved in different solvents under ambient conditions (except for DESs AACC and TGCC, where the spectra are acquired at 328.15 K). Error in $\lambda_{abs}^{max} \leq \pm 0.5$ nm and $\lambda_{flu}^{max} \leq \pm 1$ nm.

Solvent System	λ_{abs}^{max} (nm)	λ_{flu}^{max} (nm) (λ_{exc} in nm)
Water:		
pH = 2	300, 370	450 (300), 450 (370)
pH = 7	337, 349	450 (340), 450 (349)
pH = 12	337, 347	450, 380 (340)
DESs:		
URCC	337, 350	380 (340)
BDCC	340, 350	439, 380, 365 (340) 438, 380, 365 (350)
AACC	340, 350	452, 373 (340) 455, 370 (350)
EGCC	300, 350, 370	451 (300) 437, 385 (340) 382, 449 (350), 451 (370)
GLCC	300, 350, 370	450 (300), 452 (340) 450 (350), 450 (370)
TGCC	367	449 (300), 447 (340) 447 (350), 447 (370)
PACC	378	452 (300), 453 (340) 449 (350), 448 (370)
MACC	300, 374	446 (300), 439 (340) 438 (350), 442 (370)
HBDs:		
Glycerol	300, 370	450 (300), 450 (340) 450 (370)
Tetraethylene Glycol	300, 340, 350, 370	369, 380, 450 (340) 449 (370)
Ethylene Glycol	337, 350, 370	369, 380, 450 (340) 447 (370)
1,4-Butane-diol	337, 350	369, 380 (340)

In order to gain further insight to this behavior, UV–vis absorption spectra of norharmane are acquired in glycerol, ethylene glycol, 1,4-butanediol, and tetraethylene glycol - the H-bond donors that are in liquid state under ambient conditions and are involved in the formation of four of the DESs investigated (the spectra are presented in Fig. 2, panel A and corresponding absorbance maxima are reported in Table 1).

Absorption spectral characteristics of norharmane in glycerol are fairly similar to those in DES GLCC; in ethylene glycol, the band corresponding to cation is a little less prominent as compared to that in DES EGCC. Similarities in absorption spectral features within H-bond donor 1,4-butanediol and corresponding DES BDCC are stark. While the aforementioned H-bond donors control the ground state prototropic forms of norharmane in their respective ChCl : HBD DESs, for HBD TEG, it is not the same – while only the cationic form exists in DES TGCC, a careful examination of the data reveals prominent presence of neutral form of norharmane in neat TEG along with the cationic form. As TEG possess ethoxy functionalities, in order to gain further insight, norharmane absorption spectra is acquired from its solutions in liquid PEG200, PEG400, and PEG600, respectively (Fig. S2, panel A). It is interesting to note that while spectra in PEG200 and PEG400 exhibit presence of both neutral and cationic forms, no cationic form is present in PEG600. Therefore, the relative concentration of ethoxy-to-hydroxy functionalities do appear to control norharmane prototropism. Any generalization that HBD of a DES completely controls the ground state prototropic behavior of the probes may not be correct.

3.2. Steady-state fluorescence data

Upon excitation at 340 nm, in aqueous media, a broad fluorescence emission band centered at 450 nm is the prominent emission band at pH 2, 7, and 12; at pH 12, in addition a shoulder is observed at 380 nm (Fig. S1, panel B). The band at 450 nm is assigned to the excited cationic form, while the shoulder at 380 nm is to the excited neutral form. This suggests increased ease of protonation in the excited-state even at relatively higher pH values as, in the excited state, cation is the dominant form even at pH 7; excited neutral form starts to show up at pH 12 (the pK_a* ≈ 13.0). It has been reported that the zwitterionic form of norharmane can also be seen in strongly alkaline solutions (pH = 14) by the appearance of a red-shifted fluorescence band at ~510 nm [43]. The emission transitions of norharmane in aqueous solutions at different pH are listed in Table 1 and are in agreement with those reported earlier [43].

The highly efficient protonation of norharmane in the excited-state is amply reflected via the fluorescence emission spectra of this probe in the DESs investigated as well (Fig. 3 presents the emission spectra and corresponding emission maxima are reported in Table 1).

It is easy to comprehend that in the DESs, TGCC, PACC, and MACC, where only the cationic form existed in the ground state, in excited-

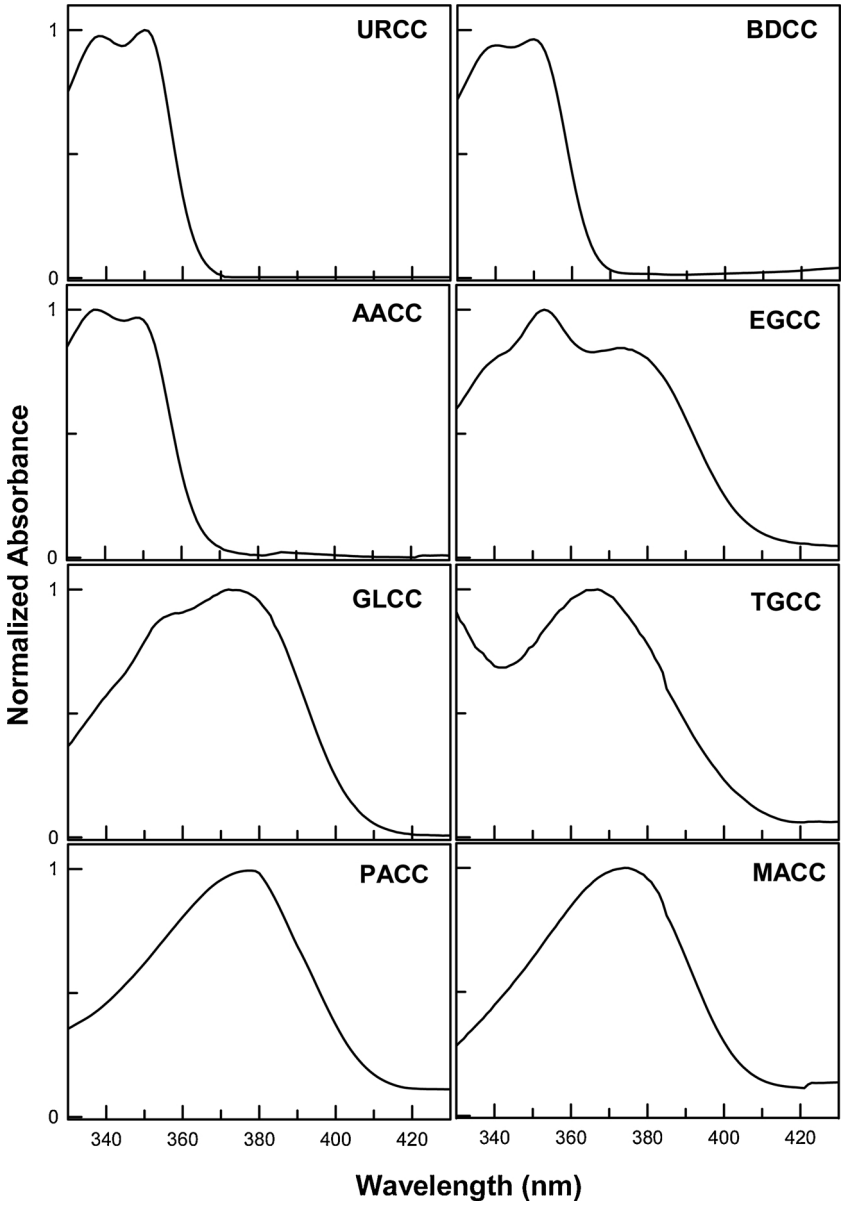


Fig. 1. Normalized UV–vis absorption spectra of norharmane (25 μM) dissolved in different DESs under ambient conditions (except for AACC and TGCC, where the spectra are acquired at 328.15 K).

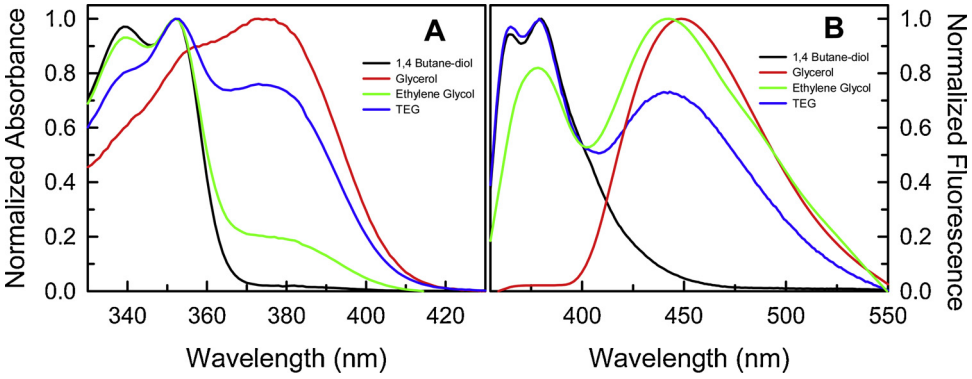


Fig. 2. Normalized UV–vis absorption (panel A) and fluorescence emission (panel B; λ_{exc} = 340 nm) spectra of norharmane (25 μM for absorption and 10 μM for fluorescence) dissolved in liquid HBD components of DESs investigated under ambient conditions.

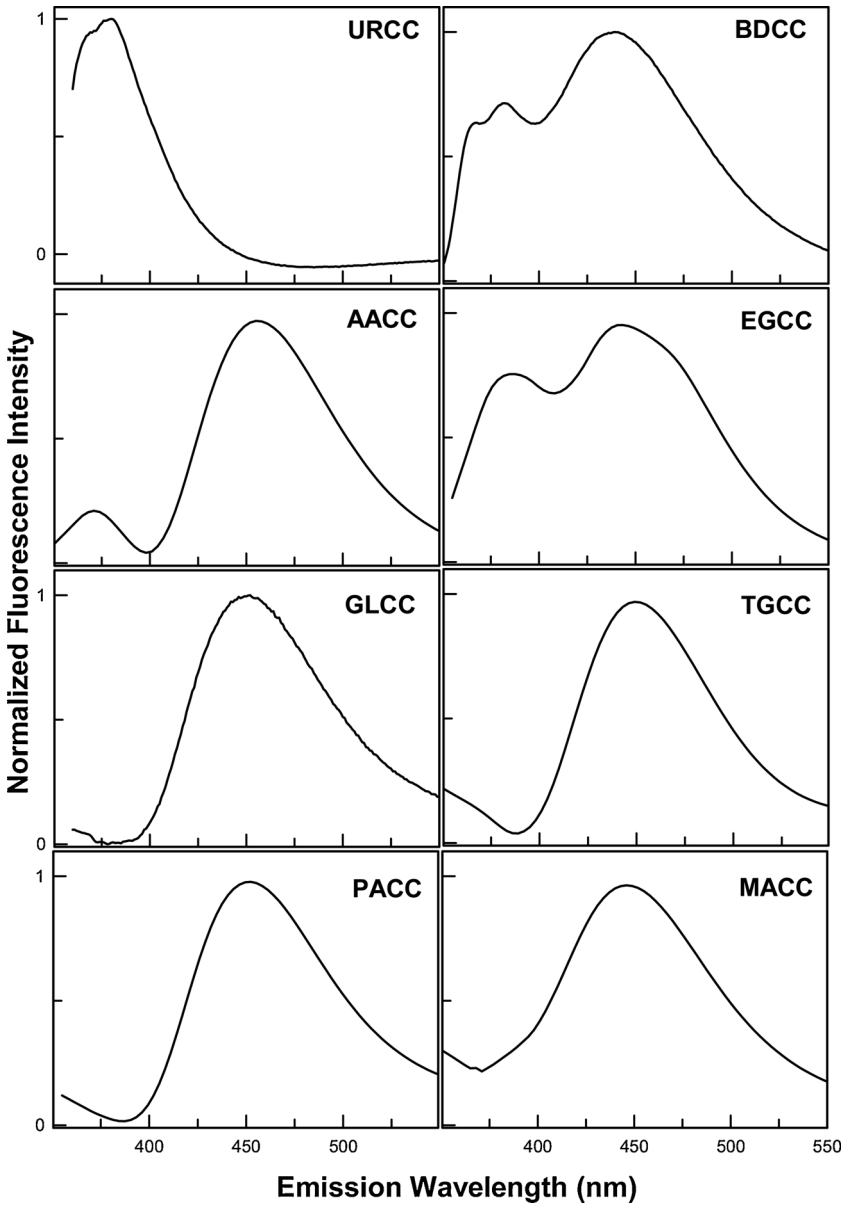


Fig. 3. Normalized fluorescence emission spectra of norharmane (10 μM, λ_{exc} = 340 nm) dissolved in different DESs under ambient conditions (except for AACC and TGCC, where the spectra are acquired at 328.15 K). Band in the region 365–380 nm is assigned to neutral form and that in the region 437–452 nm to the cationic form.

state also only the cationic form is present (broad band centered ~450 nm). Emission spectra in GLCC confirm the presence of only the cationic form in the excited state; in EGCC, along with the cationic form, presence of neutral form in the excited state is also evident by the additional peak 382 nm. In AACC and BDCC, where only the neutral form exists in the ground state, cationic form becomes dominant in the excited state. URCC is the one DES among the eight DESs investigated where only the neutral form of norharmane exists in the excited state which is characterized by the emission band centered at 380 nm. It is established that the prototropism of norharmane in the excited state within a DES is dependent on the identity of the HBD used to prepare the DES.

The fluorescence emission of norharmane is collected in HBDs glycerol, ethylene glycol, 1,4-butanediol, and tetraethylene glycol (Fig. 2, panel B presents fluorescence emission spectra with emission maxima reported in Table 1); these are the only HBDs of the DESs investigated in this work that are in liquid state under ambient conditions. Fluorescence emission of norharmane in HBDs glycerol and ethylene glycol closely resemble those in the corresponding DESs GLCC and EGCC, respectively. While only excited cationic form of norharmane exists in DES TGCC, band corresponding to the neutral form is dominant in neat TEG. It is interesting to note that while both excited cationic and neutral forms exist in DES BDCC, only excited neutral form exists in corresponding neat HBD 1,4-butanediol. It is apparent that DESs support cationic form more than the corresponding HBDs that they are constituted of.

To establish structural correlation, if any, absorption and fluorescence emission spectra of norharmane are also acquired in various other organic solvents (spectra are depicted in Fig. S3 and band maxima are reported in Table S1). The data in the investigated organic solvents reveal that within *n*-alkyl alcohols, neutral form is the dominant form in the ground state (represented through 350 nm absorbance band). While shorter alkyl chain alcohols (methanol, ethanol) do also support cationic form to some extent (represented via shoulder at 370 nm), as the alkyl chain becomes longer the cationic form becomes less and less. However, 2,2,2-trifluoroethanol (TFE) offers considerably improved

presence of the cationic form. Fluorescence emission spectra in TFE clearly reveals the presence of only the cationic form in the excited state; except for methanol where a shoulder characterizing the cationic form appears, excited cation of norharmane does not exist in any other investigated alkyl alcohol, only the neutral form exists in the exited state in these alcohols. It may be inferred that the considerably higher HBD acidity (one way to manifest HBD acidity is via Kamlet-Taft HBD acidity parameter, α, reported in Table S3) of TFE (and of methanol to some extent) as compared to other alkyl alcohols facilitates proton donation thus helping the cationic form to exist in the solution. This is further confirmed by the complete absence of the cationic form in both ground and excited states in a highly polar but aprotic solvent DMSO.

3.3. Excited-state emission intensity decay

Excited-state proton transfer dynamics of norharmane has been investigated in various organic solvents and in aqueous media at different pH. Apart from polar-protic methanol (and acidified methanol) and ethanol, the excited-state emission intensity decays have been collected in polar-aprotic acetonitrile as well. In a detailed investigation of norharmane proton transfer dynamics, Reyman group collected and interpreted norharmane excited state intensity decay in up to 20% acetic acid solutions of dichloromethane [36]. Norharmane emission intensity decay were also reported in other halogenated solvents, such as, chloroform, carbon tetrachloride, and dibromomethane along with nonpolar toluene [38,44]. Lack of agreement among the researchers on recovered decay times of various prototropic forms of norharmane is apparent from these reports [38,47].

We first acquired the excited-state emission intensity decay of norharmane at 450 nm from its aqueous solutions having pH 2, 7, and 12 (Table S2). The decay at pH 2 fits best to a single exponential decay equation with recovered decay time of 21.7 ns characterizing the excited cationic form. This decay time is in excellent agreement with that (22 ns) reported in literature for pH < 5 solutions [38]. However, at pH 7, along with a 21.6 ns dominant component (95%), a shorted decay time of 10.3 ns could also be recovered (at pH 8–11, the reported decay time is 8 ns) [33]. We recovered a decay time of 3.1 ns at pH 12 which is in agreement with the decay times reported in methanol [47] (2.8 ns), DCM [37] (2.5 ns), and chloroform [46] (2.1 ns). This decay time could be tentatively attributed to the neutral form of norharmane.

Excited-state emission intensity decay, *I(t)*, of norharmane were collected from its solutions in the eight investigated DESs. For URCC the decay was collected at 380 nm as emission from only the neutral form of norharmane is observed in this DES. The decay exhibited best fit to a single exponential decay function with recovered decay time of 4.6 ns (decay and fit to the decay are shown in Fig. 4 and recovered parameters are reported in Table 2). This decay time is attributed to the neutral form of norharmane; it is to be noticed that the decay time is higher than that reported in organic solvents and water (~3 ns) [49]. Fluorescence emission characterizing only the excited cationic form of norharmane is observed from DESs GLCC, TGCC, PACC, and MACC, and hence, *I(t)* were collected at 450 nm from these DESs. Fig. 5 presents representative decay and fits in GLCC and recovered parameters are reported in Table 2. It is clear from the data that *I(t)* fit to a single exponential decay function is not satisfactory in these four DESs (based on goodness-of-the-fit in terms of χ^2, weighted residuals, and DW parameters); fit to a double exponential decay equation affords considerably improvement. One of the two decay times is in the range 0.4–1.8 ns (the shorter of the two decays) and the other longer one is in the range 16.8–20.0 ns. While the longer is easily assignable to the cationic form of norharmane, the origin of the shorter one is debatable in the literature and is reported to be inherent to the cationic form as well [38,47]. In many earlier publications, *I(t)* of norharmane in other solvents collected at 450 nm are reported to fit better to double or triple exponential decay functions with multiple decay times assigned to the cationic form [47,37].

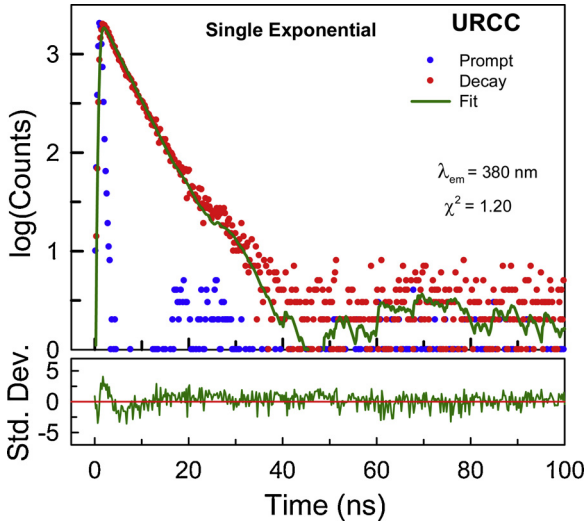


Fig. 4. Excited-state intensity decay and its fit to single exponential decay equation for norharmane (10 μM; excitation with 340 nm NanoLED; emission collected at 380 nm) dissolved in DES URCC under ambient conditions. Residuals are provided below the panel.

Table 2
Recovered excited-state intensity decay parameters for norharmane (10 μM; excitation with 340 nm NanoLED) dissolved in different DESs. Errors associated with decay times are ≤ ± 5% and with pre-exponential factors are ≤ ± 5%.

DES	λ_{em} (nm)	τ_1 (ns)	τ_2 (ns)	α_1	α_2	χ^2
URCC	380	4.6	–	1.0	–	1.20
BDCC	380	5.8	–	1.0	–	1.14
	450	6.7	–	1.0	–	2.61
		5.0	14.6	0.9	0.1	1.14
AACC	380	4.2	–	1.0	–	1.05
	450	6.0	–	1.0	–	5.33
		2.6	11.5	0.8	0.2	1.38
EGCC	380	5.4	–	1.0	–	1.21
	450	0.2	–	1.0	–	1.74
		1.7	17.3	0.2	0.8	1.50
GLCC	450	16.0	–	1.0	–	4.86
		2.0	18.0	0.6	0.4	1.21
TGCC	450	15.6	–	1.0	–	5.22
		1.4	17.6	0.7	0.3	1.20
PACC	450	18.0	–	1.0	–	5.88
		1.1	20.0	0.7	0.3	1.28
MACC	450	16.3	–	1.0	–	2.52
		0.4	16.8	0.7	0.3	1.36

The above observation is further supported by the fit of the norharmane *I(t)* within DESs (BDCC, AACC, and EGCC) where emission spectra are highlighted by the presence of the bands characterizing both neutral and cationic forms. In these DESs, *I(t)* is collected at both 380 nm and at 450 nm. It is interesting to note that within all three DESs, the decay at 380 nm fit best to a single exponential decay function with the recovered decay time in the range 4.2–5.8 ns (Fig. 6 presents representative *I(t)* and fits within the DES AACC, while the recovered parameters for the DESs are reported in Table 2). It is convenient to note that these recovered decay times at 380 nm encompass the decay time recovered for URCC (vide supra), and hence are ascribable to the neutral form of norharmane. Similar to that observed for DESs where on the emission from the cationic form of norharmane is observed, the *I(t)* at 450 nm within BDCC, AACC, and EGCC again do not fit satisfactorily to a single exponential decay function; a double exponential decay equation affords considerably better fit to the *I(t)* (Fig. 6 and Table 2). Again, the longer of the two recovered decay times is in the range 11.5–17.3 ns and the shorter is in the range 1.7–4.9 ns; as before, the two decay times are attributed to the cationic form of

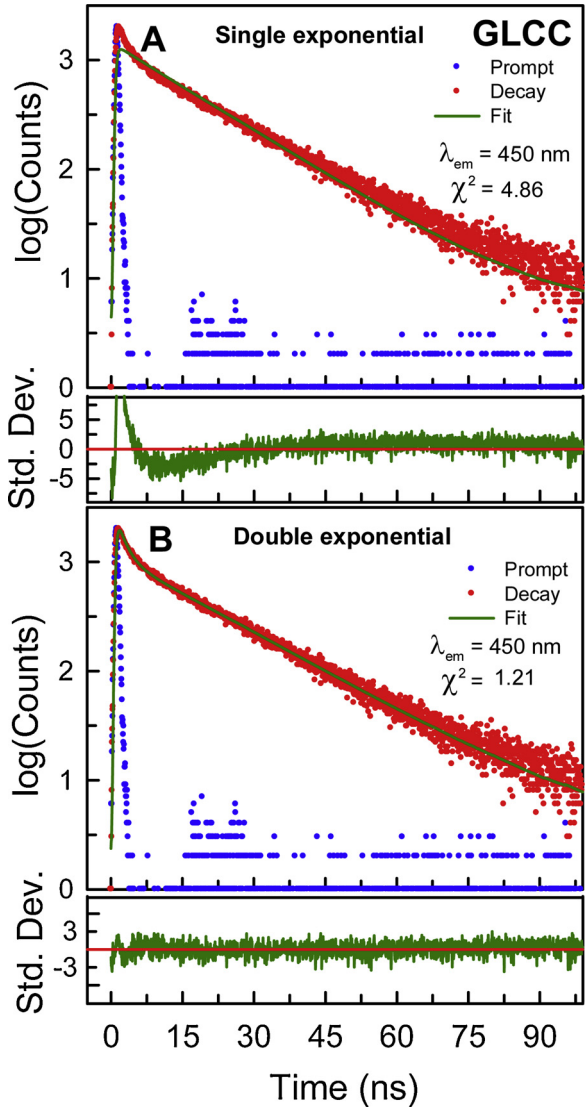


Fig. 5. Excited-state intensity decay data fit to single exponential (panel A) and double exponential (panel B) decay equation of norharmane (10 μM; excitation with 340 nm NanoLED; emission collected at 450 nm) dissolved in DES GLCC under ambient conditions. Residuals are provided below each panel.

norharmane. The excited-state intensity decay of norharmane within the eight DESs investigated amply complement the steady-state fluorescence emission data and asserts that the existence of a prototropic form depends on the HBD constituting the DES, and in turn, support the dependence on the HBD acidity and HBA basicity of these DESs.

3.4. Prototropic forms and E_T^N/Kamlet-Taft parameters

The possible role of HBD acidity parameter on prototropic forms is mentioned during the discussion of relative presence of neutral and cationic forms of norharmane in other organic solvents in the previous section. Subsequently, we have assessed the empirical solvent parameter E_T^N of solvent polarity along with Kamlet-Taft parameters – dipolarity/polarizability (π^*), HBD acidity (α), and H-bond accepting basicity (β) – of six of the eight DESs investigated (Table 3). Due to highly acidic nature of PACC and MACC, these parameters could not be obtained for these two DESs as no lowest energy electronic absorbance transition could be observed for either the Reichardt's dye or betaine dye 33 that is needed to assess E_T^N and α. This is due to the extensive protonation of these dyes. Our values of the DESs that are already reported in literature are in good agreement [66].

It is convenient to notice that the DES URCC which supports exclusively the neutral form of norharmane in both ground and excited states has the lowest E_T^N (0.62) of the DESs investigated. As the contribution from H-bond donating acidity towards E_T^N is ca. 68%, the HBD acidity, alpha, is also the lowest for URCC (α = 0.39). Thus, very weak HBD acidity of URCC is amply manifested in the fact that conversion from neutral to cationic form of norharmane is not facilitated at all in this DES. Understandably, DES AACC in which only neutral form exists in ground state has the second lowest HBD acidity among the group; the higher HBD acidity of AACC as compared to that of URCC (α = 0.65 versus 0.39), however, results in appreciable formation of cationic form in the excited state. The HBD acidity of DES BDCC is fairly high, however, it also has the highest HBA basicity among the DESs investigated. It appears the high HBA basicity dominates norharmane prototropism in the ground state allowing only the neutral form to exist, while along with the neutral form the cationic form appears in the excited state due to appreciable HBD acidity of this DES (α = 0.84). Within GLCC, the DES having the highest of the HBD acidities (α = 0.94), only the cationic form is supported in the excited state; however, due to appreciable HBA basicity (β = 0.49), small amount of neutral form also exists in the ground state. Conveniently, within the DESs constituted of highly acidic HBDs, PACC and MACC, only the cationic form of norharmane is exclusively present in both ground and excited states. It is inferred that the prototropism of norharmane within DESs is controlled by a complex interplay of both HBD acidity and HBA basicity of the media.

The presence of norharmane prototropic forms and its correlation with Kamlet-Taft parameters is fairly apparent for organic solvents (Table S3 and Fig. S3). In DMSO, a solvent with no HBD acidity but appreciable HBA basicity, only the neutral form exists in both ground and excited states. For TFE, water, and glycerol - solvents with very high HBD acidity - only form in the excited state is the cationic form. While the high HBD acidity combined with moderate HBA basicity of methanol helps support some cationic form in the excited state, relatively lower HBD acidity is dominated by the higher HBA basicity within *n*-butanol, *n*-propanol, and 1,4-butanediol giving rise to only the neutral form in the excited state. Solvents with moderate HBD acidity as well as HBA basicity tend to support both cationic and neutral forms in the excited state (EG and TEG).

4. Conclusions

The role of HBD constituent of DESs on prototropic behavior of a model probe norharmane is assessed via investigation of eight structurally-different HBD-containing DESs. In the ground-state, only the neutral form of norharmane exists in DESs URCC (due to very low HBD acidity), BDCC (due to high HBA basicity), and AACC (due to relatively lower HBD acidity and higher HBA basicity). In GLCC and EGCC, high HBD acidity leads to the appearance of the cationic form along with the neutral form. In highly acidic DESs PACC and MACC and in DES TGCC, only the cationic form of norharmane is present. As cationic form is the more preferred form in the excited-state as compared to the ground-state, DESs PACC, MACC and TGCC support only the cationic form in the excited-state. In GLCC also only the cationic form is present in the excited state due perhaps to its highest HBD acidity among all DESs investigated. Along with EGCC, the cationic form also appears in the excited-state within DESs AACC and BDCC. Due to the lowest HBD acidity among all DESs investigated, URCC is the one DES where only the neutral form of norharmane exists in the excited state. Based on the prototropism in neat liquid HBDs, one should refrain from any generalization that the prototropic behavior of the probe within a DES is only controlled by the constituent HBD; subtle differences may exist between the HBD and the DES composed of it. DESs as novel, alternate, non-toxic and inexpensive media to modulate prototropic behavior of probes depending on the constituents of the DES is established in this work.

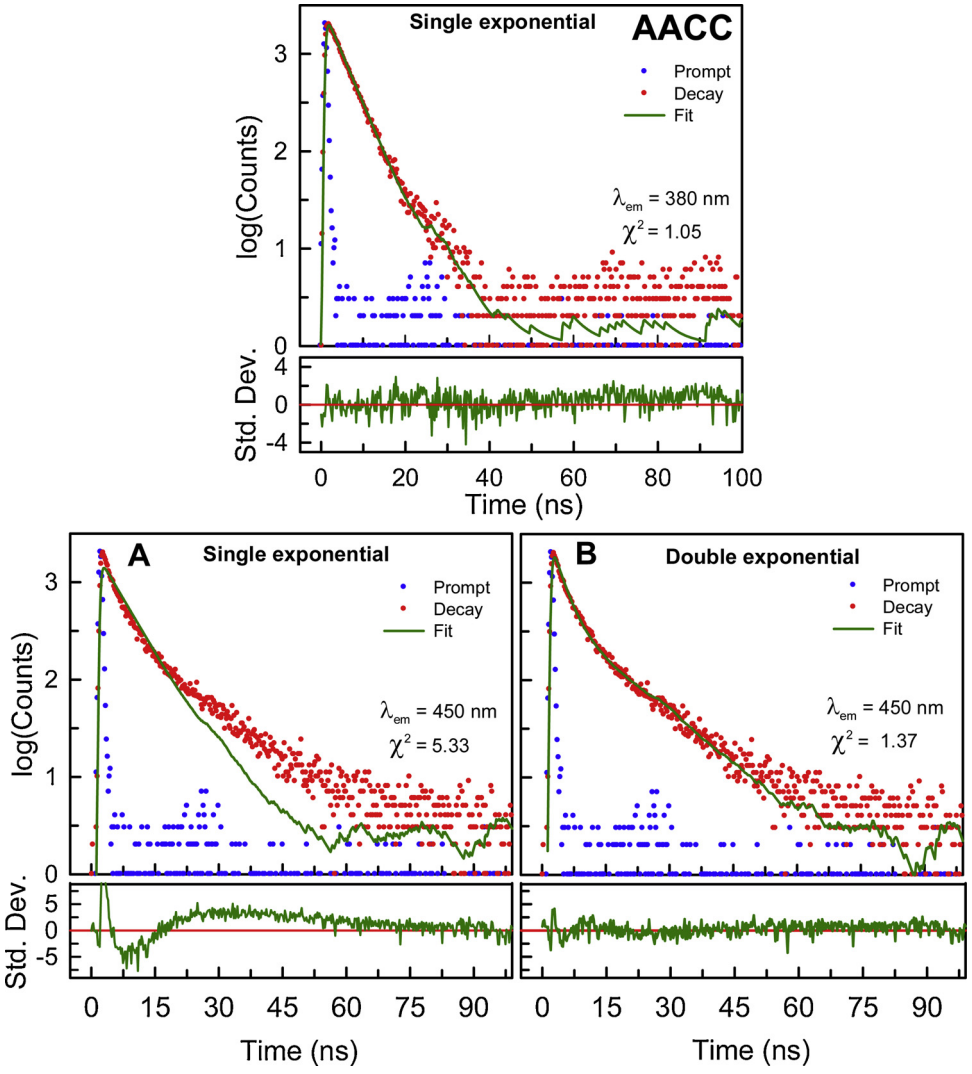


Fig. 6. Representative plots showing fits of excited-state intensity decay of norharmane (10 μM) to single-exponential decay function at λ_{em} = 380 nm (top panel), single-exponential decay function at λ_{em} = 450 nm (bottom panel A) and double exponential function λ_{em} = 450 nm (bottom panel B) dissolved in DES AACC at 328.15 K. Excitation was carried out using 340 nm LED. Residuals are provided below each panel.

Table 3
E_T^N and Kamlet-Taft Parameters - α (HBD acidity) and β (HBA basicity) - of different DESs used in this study and the norharmane species obtained in these solvents. Errors associated to $E_T^N \leq \pm 0.02$, $\alpha \leq \pm 0.02$ and $\beta \leq \pm 0.02$.

DES	E_T^N	α	β	Norharmane Species	
				Ground State	Excited State
URCC	0.62	0.39	0.36	N	N*
AACC	0.70	0.65	0.58	N	N* + C*
TGCC	0.72	0.69	0.50	C	C*
BDCC	0.78	0.84	0.62	N	N* + C*
EGCC	0.84	0.91	0.46	N + C	N* + C*
GLCC	0.87	0.94	0.49	N + C	C*

Where, N = Neutral form of norharmane, C = Cationic form of norharmane, and * is used to represent the excited state.

Declaration of Competing Interest

Authors declare no conflict of interest.

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Appendix A. Supplementary data

Supplementary material related to this article can be found, in the online version, at doi:https://doi.org/10.1016/j.jphotochem.2019.112138.

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